

Prophet: Marketing Intelligence & Incrementality

Technical and Executive Report

December 28, 2025

1 Executive Summary

The **Prophet Marketing Intelligence Platform** is an end-to-end analytics and modeling system designed to **predict, measure, and causally attribute customer behavior** in response to marketing interventions such as advertisements and offers.

Traditional marketing analytics often rely on correlation-based metrics (click-through rate, last-touch attribution, raw conversion rates). Prophet moves beyond correlation by combining:

- Predictive modeling (purchase likelihood, churn, customer lifetime value)
- Causal inference and uplift modeling
- Executive-grade visualization and decision support

The system answers a central business question:

Did our marketing campaigns create incremental value, and if so, for which customers?

Empirical results show that advertising exposure leads to **statistically significant and economically meaningful uplift** in both purchase probability and customer lifetime value when applied selectively rather than uniformly.

2 System Architecture Overview

Prophet is implemented as a modular, production-style analytics stack composed of four layers:

2.1 Event and Feature Layer

- Source data includes user-level behavioral events (views, searches, purchases, ad exposure)
- Time-windowed aggregations (1 hour to 180 days)
- Momentum, recency decay, burstiness, and engagement intensity features
- Strict leakage control using anchor-time-based feature construction

2.2 Predictive Modeling Layer

- Purchase Likelihood: 1d, 7d, 30d, 90d horizons
- Churn Prediction: 30-day horizon
- Customer Lifetime Value (CLV): 30d and 90d spend regression
- Models: LightGBM, CatBoost, and stacked ensembles

2.3 Causal Inference Layer

- Treatment definition: advertising exposure
- Implemented learners:
 - T-Learner
 - S-Learner
 - X-Learner (propensity-adjusted)
- Outputs individual treatment effects (ITE), average treatment effects (ATE), and policy simulations

2.4 Visualization and Decision Layer

- Streamlit dashboards for executives and engineers
- Model performance, uplift visualization, and policy insights
- Designed for live walkthroughs and strategic decision-making

3 Predictive Modeling Performance

3.1 Purchase Likelihood Models

Purchase likelihood models were trained across multiple horizons using gradient-boosted decision trees.

Horizon	ROC-AUC	PR-AUC	Precision@Top 5%
1 Day	0.8506	0.1830	0.224
7 Days	0.8941	0.6627	0.806
30 Days	0.8172	0.9569	–
90 Days	0.9586	0.9982	–

Key observations:

- Short-term purchases are noisy and highly imbalanced
- Predictive signal strengthens over medium horizons
- Long-term purchase intent is highly separable
- Top-decile precision validates real-world targeting use

3.2 Churn Prediction (30 Days)

The churn model achieved strong discriminative performance:

- ROC-AUC: ~ 0.90
- PR-AUC: ~ 0.91
- Precision@Top 5%: 100%

Primary churn drivers include recency, engagement decay, session frequency, and advertising interaction patterns. The model enables proactive intervention rather than reactive retention.

3.3 Customer Lifetime Value (CLV)

A stacked ensemble combining LightGBM and CatBoost achieved:

- $R^2 = 0.97$
- RMSE ≈ 11.6
- Top-decile lift: $5.4\times$

This confirms strong revenue concentration among a small subset of customers.

4 Causal Inference and Uplift Modeling

Predictive accuracy alone cannot establish causality. Prophet implements uplift modeling to estimate incremental impact.

4.1 Treatment Definition

- Treatment group: customers exposed to ads
- Control group: customers not exposed
- Propensity modeling used to adjust for selection bias

4.2 Uplift Results (CLV)

Method	ATE	ATT
T-Learner	+0.97	-0.75
S-Learner	0.00	0.00
X-Learner	+0.70	+0.81

The X-Learner provides the most stable and interpretable estimates, capturing heterogeneous treatment effects.

4.3 Policy Simulation

Using the top-decile uplift segment identified by the X-Learner:

- Average CLV (treated): 87.18
- Average CLV (control): 61.22
- Incremental lift: $\sim 42\%$

This demonstrates that targeted advertising materially outperforms blanket exposure.

5 Executive Implications

Prophet enables a shift from heuristic-driven marketing to evidence-based decision-making:

- Budget allocation based on expected incremental value
- Suppression of non-incremental users
- CLV-weighted campaign prioritization
- Proactive churn mitigation

Marketing becomes a measurable growth engine rather than a cost center.

6 Future Extensions

Planned enhancements include:

- Offer- and creative-level uplift modeling
- Real-time decision engines
- Budget-to-revenue simulation tools
- Continuous learning from campaign feedback

7 Conclusion

Prophet demonstrates how modern machine learning and causal inference can be combined to deliver actionable, trustworthy marketing insights. By separating prediction from causation, the platform provides clarity on what truly drives incremental business value.